
Hands-On Lab: AWS Trusted Advisor

Objective

Use **AWS Trusted Advisor** to check for **best practice violations** in your AWS environment and learn how to take corrective action (e.g., close open security ports, delete unused resources).

Prerequisites

- AWS Account (with **Business or Enterprise** support plan for full access, or **Basic** for core checks)
 - A few AWS resources to analyze (e.g., EC2, S3, IAM, EBS)
 - IAM permissions:
 - `support:DescribeTrustedAdvisorChecks`
 - `support:DescribeTrustedAdvisorCheckResult`
-

Step-by-Step Instructions

♦ Step 1: Access Trusted Advisor

1. Sign in to the [AWS Trusted Advisor Console](#)
2. If prompted, click “**Go to full Trusted Advisor**” for Business or Enterprise plan
3. You'll see a dashboard with checks grouped by:

-  Cost Optimization
 -  Security
 -  Performance
 -  Fault Tolerance
 -  Service Limits
-

Step 2: Review Check Results

View Cost Optimization

- Look for checks like:
 - **Idle EC2 Instances**
 - **Underutilized EBS Volumes**
- Click into a check to view:
 - Affected resources
 - Cost impact
 - Suggested action

View Security Recommendations

- Key checks:
 - **MFA on Root Account**

- IAM Access Key Rotation
- S3 Bucket Permissions
- Fix issues directly (e.g., apply bucket policies, rotate keys)

Check Service Limits

- Look for services approaching 80% usage (e.g., EC2 Instances, VPCs)
- You can request quota increases from the same screen

Step 3: Take Action on One Issue

1. Choose a **security group with open ports** (e.g., 0.0.0.0/0 on port 22 or 3389)
2. Go to **EC2 → Security Groups**
3. Edit **inbound rules** to restrict IPs or remove the open rule

 Wait a few minutes and refresh Trusted Advisor — the issue should move to “Green” 

Optional: Use AWS CLI to Query Checks

List available checks:

```
aws support describe-trusted-advisor-checks --language en
```

1.

Get check ID for **MFA on Root Account** (for example):

```
aws support describe-trusted-advisor-checks --language en \
| jq '.checks[] | select(.name=="MFA on Root Account") | .id'
```

2.

Get the result:

```
aws support describe-trusted-advisor-check-result \  
--check-id <CHECK-ID> \  
--language en
```

3.

✅ Step 4: Set Up Weekly Alerts (Optional)

To get alerts for Trusted Advisor changes:

1. Go to **CloudWatch** → **Rules**
2. Create a rule for **event pattern**:
 - Service: **TrustedAdvisor**
3. Set **target** as an **SNS topic**
4. Subscribe your email to the SNS topic

📧 You'll now receive weekly updates on check status changes.

🧹 Cleanup (if any test resources were created)

- Terminate unused EC2 instances
- Delete test EBS volumes
- Secure S3 buckets

- Remove unnecessary IAM users or access keys

Summary

Task	Completed
Accessed Trusted Advisor Console	☒
Reviewed 5 core checks	☒
Fixed at least 1 issue (e.g., open port, idle resource)	☒
(Optional) Queried via CLI	☒
(Optional) Set up alerting	☒
